

Patient-Specific Multi-Omics Graph Transformer

Architecture & Mathematical Formulation Reference

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Abstract

This document serves as an exhaustive reference for the Patient-Specific Multi-Omics Graph Transformer project. It details the transition from static, message-passing Graph Convolutional Networks (GCNs) to a globally attentive Graph Transformer. By leveraging hierarchical attention (intra-gene and inter-gene) and injecting Protein-Protein Interaction (PPI) networks as structural priors, the model maps complex, non-linear epigenetic and transcriptomic interactions. This specification covers the rigorous pre-processing pipeline, the complete mathematical formulation of the architecture, regularization strategies (Modality Dropout, DropEdge), and the native interpretability framework.

1 Data Preprocessing & Pipeline Alignment

Rigorous preprocessing is critical to prevent data leakage and ensure scale uniformity across diverse biological modalities (mRNA, CNV, Methylation).

1.1 Alignment and Splitting

Let the raw inputs be $\mathbf{X}_{mRNA}$, $\mathbf{X}_{CNV}$, and $\mathbf{X}_{methy}$, alongside a clinical vector $\mathbf{Y}$ containing subtype labels (e.g., Luminal A, Basal).

1. **Patient Intersection:** Only patients existing across all three modalities and the clinical dataset are retained.
2. **Train/Validation Split:** A stratified split is performed *prior* to any statistical calculation to prevent data leakage.

1.2 Highly Variable Gene (HVG) Selection via MAD

To extract the $N = 2000$ most informative genes while remaining robust to biological outliers, we utilize the Median Absolute Deviation (MAD), fitted strictly on the training set:

$$\text{MAD}(\mathbf{x}) = \text{median}(|\mathbf{x} - \text{median}(\mathbf{x})|) \quad (1)$$

The genes are ranked independently by their MAD scores for each modality, and the consensus rank is calculated as:

$$\text{Rank}_{consensus} = \text{Rank}_{mRNA} + \text{Rank}_{CNV} + \text{Rank}_{methy} \quad (2)$$

The top N genes with the lowest $\text{Rank}_{consensus}$ are selected.

1.3 Modality-Specific Normalization

Normalization parameters are fitted on the training split and transformed globally:

- **RNA-seq:** $\mathbf{x}'_{mRNA} = \text{StandardScale}(\ln(\mathbf{x}_{mRNA} + 1))$
- **CNV & Methylation:** $\mathbf{x}' = \text{StandardScale}(\mathbf{x})$

2 Architectural Formulation

The model architecture is hierarchical, proceeding from raw scalar values to dense modality embeddings, then to unified gene tokens via local attention, and finally to a holistic patient state via global graph attention.

2.1 Modality Lifting & Biological Embeddings

For each patient and for each selected gene $i \in \{1, \dots, N\}$, the scalar values $x_{m,i}, x_{c,i}, x_{t,i}$ are projected into a continuous, d -dimensional space ($d = 64$). Learnable modality embeddings $\mathbf{E}_m, \mathbf{E}_c, \mathbf{E}_t \in \mathbb{R}^d$ are added to encode the biological origin of the signal.

$$\mathbf{z}_{m,i} = \text{Linear}_{mRNA}(x_{m,i}) + \mathbf{E}_m = (x_{m,i} \mathbf{W}_m + \mathbf{b}_m) + \mathbf{E}_m \quad (3)$$

$$\mathbf{z}_{c,i} = \text{Linear}_{CNV}(x_{c,i}) + \mathbf{E}_c = (x_{c,i} \mathbf{W}_c + \mathbf{b}_c) + \mathbf{E}_c \quad (4)$$

$$\mathbf{z}_{t,i} = \text{Linear}_{methy}(x_{t,i}) + \mathbf{E}_t = (x_{t,i} \mathbf{W}_t + \mathbf{b}_t) + \mathbf{E}_t \quad (5)$$

where $\mathbf{W} \in \mathbb{R}^{1 \times d}$ and $\mathbf{b} \in \mathbb{R}^d$.

2.2 Mini-Transformer: Intra-Gene Fusion

To model local, non-linear epigenetic and transcriptomic interactions (e.g., methylation silencing of mRNA expression), we employ a local cross-attention mechanism inside each gene.

We formulate the input sequence for gene i by prepending a learnable gene-level classification token $\mathbf{z}_{cls} \in \mathbb{R}^d$:

$$\mathbf{X}_i = \begin{bmatrix} \mathbf{z}_{cls} \\ \mathbf{z}_{m,i} \\ \mathbf{z}_{c,i} \\ \mathbf{z}_{t,i} \end{bmatrix} \in \mathbb{R}^{4 \times d} \quad (6)$$

Modality Dropout Regularization: During training, with probability $p_{drop} \in [0.1, 0.2]$, the mRNA embedding is masked:

$$\tilde{\mathbf{z}}_{m,i} = \begin{cases} \mathbf{0}, & \text{with probability } p_{drop} \\ \mathbf{z}_{m,i}, & \text{otherwise} \end{cases} \quad (7)$$

This forces the network to utilize epigenetic profiles rather than overfitting to dominant mRNA signals.

Intra-Gene Attention: We apply multi-head self-attention (MHA). Using the Pre-LayerNorm formulation:

$$\mathbf{X}'_i = \mathbf{X}_i + \text{MHA}(\text{LN}(\mathbf{X}_i)) \quad (8)$$

$$\mathbf{X}''_i = \mathbf{X}'_i + \text{MLP}(\text{LN}(\mathbf{X}'_i)) \quad (9)$$

where the attention operation for a single head is defined as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} \right) \mathbf{V} \quad (10)$$

Token Extraction: The unified biological representation of gene i , denoted $\mathbf{h}_i$, is extracted as the updated state of the classification token:

$$\mathbf{h}_i = \mathbf{X}_{i,0}'' \in \mathbb{R}^d \quad (11)$$

2.3 Structural Injection: Graph Positional Encodings

To provide the Transformer with normal biological proximity priors without restricting information flow to rigid multi-hop message passing, we inject Graph Positional Encodings derived from the STRINGdb PPI network.

Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ be the adjacency matrix of the base PPI network. The degree matrix is $\mathbf{D} = \text{diag}(\mathbf{A}\mathbf{1})$.

2.3.1 Dynamic DropEdge Regularization

During training, edges are randomly dropped at each epoch with probability $q_{drop} \approx 0.10$ to prevent overfitting to hub genes:

$$\tilde{\mathbf{A}}_{jk} = \begin{cases} 0, & \text{with probability } q_{drop} \\ \mathbf{A}_{jk}, & \text{otherwise} \end{cases} \quad (12)$$

2.3.2 Random Walk Positional Encodings (RWPE)

RWPE provides a GPU-efficient metric of structural context. The transition matrix is $\mathbf{P} = \mathbf{D}^{-1}\tilde{\mathbf{A}}$. We track the return probability of a random walker for K steps ($K = 16$). The encoding for node i is:

$$\mathbf{e}_{graph,i} = [\mathbf{P}_{ii}, (\mathbf{P}^2)_{ii}, \dots, (\mathbf{P}^K)_{ii}] \in \mathbb{R}^K \quad (13)$$

2.4 Global Graph Transformer: Inter-Gene Network

The extracted unified gene tokens $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_N]$ are enriched with the structural priors. The K -dimensional RWPE is projected to dimension d and added directly:

$$\tilde{\mathbf{h}}_i = \mathbf{h}_i + \text{Linear}_{PE}(\mathbf{e}_{graph,i}) \quad (14)$$

The Global Sponge: A master token $\mathbf{t}_{cls} \in \mathbb{R}^d$ is prepended to the sequence to aggregate systemic, trans-regulatory deregulation:

$$\mathbf{H}^{(0)} = [\mathbf{t}_{cls}, \tilde{\mathbf{h}}_1, \tilde{\mathbf{h}}_2, \dots, \tilde{\mathbf{h}}_N] \in \mathbb{R}^{(N+1) \times d} \quad (15)$$

This sequence is passed through L layers of a Transformer Encoder. To solve the $\mathcal{O}(N^2)$ memory bottleneck for $N = 2000$, we utilize FlashAttention (exact linear-memory attention):

$$\mathbf{H}^{(L)} = \text{TransformerEncoder}(\mathbf{H}^{(0)}) \quad (16)$$

2.5 Classification Head

The final systemic state of the patient is captured in the updated master token:

$$\mathbf{u}_{tumor} = \mathbf{H}_0^{(L)} \in \mathbb{R}^d \quad (17)$$

Subtype probabilities are generated via an MLP:

$$\hat{\mathbf{y}} = \mathbf{W}_2(\text{GELU}(\text{BatchNorm}(\mathbf{W}_1 \mathbf{u}_{tumor} + \mathbf{b}_1))) + \mathbf{b}_2 \quad (18)$$

Training minimizes the standard Cross-Entropy Loss, while the primary evaluation metric is the **Macro F1-Score** to account for heavy class imbalances (e.g., Luminal A vs. Basal).

3 Native Interpretability Strategy

The architecture inherently supports interpretability without external algorithms like LRP:

1. **Global Driver Extraction:** Extract the attention weights of $\mathbf{t}_{cls}$ from the final global layer to identify the top K driver genes heavily attended to by the model.
2. **Deep Drill-Down:** For the identified driver genes, inspect the internal 4×4 attention matrix of their respective Mini-Transformer to isolate which omics modality (mRNA, CNV, or Methylation) drove the anomalous gene token state.